Thibault Meunier

thibmeu.com · github.com/thibmeu · cv@thibault.uk

Principal Research Engineer at Cloudflare working on privacy-preserving authentication for the web.

Experience

Cloudflare — London, then Remote · May 2019 – present
Principal Research Engineer (2025–) · Research Engineer (2020–2025) · Systems Engineer (2019–2020)

- Lead research on Privacy Pass and bot authentication; upgraded Cloudflare's deployment to the IETF standard, the protocol behind Private Access Tokens. It serves billions of requests a day.
- Initiated Web Bot Auth, cryptographic verification of bot and agent traffic with HTTP message signatures. Major CDN, cloud, and AI providers now run independent implementations.
- Built anonymous-credential rate limiting for bots and agents without compromising user privacy.
- Shipped key transparency auditing for WhatsApp's end-to-end encrypted messaging: public dashboard, Real World Crypto talk, and follow-up standardisation work.
- Co-designed the Cryptographic Attestation of Personhood, a CAPTCHA alternative studied at SOUPS 2022.
- Designed and implemented lock-free probabilistic caching in Cloudflare's CDN.
- Mentor team members, and have written more than 15 posts for the Cloudflare blog.

LiquidChain GmbH — Zug, Switzerland · Mar 2018 – Apr 2019
Research Engineer — distributed-ledger research and prototyping.

CERN — Geneva, Switzerland · Jun – Sep 2017
Intern — built a developer-tools web portal in React.

Isode Ltd — Hampton, UK · Jun – Sep 2016
Software Engineering Intern — built a Unicode-compliant emoji keyboard and TCL-based testing for XMPP.

Standards & service

- Chair of the IETF Open Cloud Mesh (OCM) working group, which standardises federated file sharing.
- Batched Token Issuance Protocol, submitted for publication.
- Privacy Pass Reverse Flow, an extension of the Privacy Pass issuance flow.
- Author of the Web Bot Auth drafts (protocol, registry).

Selected publications & writing

- Let The Right One In: Attestation as a Usable CAPTCHA Alternative — SOUPS 2022.
- Support for modern cryptographic algorithms in Workers — Cloudflare blog, 2026.
- Forget IPs: using cryptography to verify bot and agent traffic — Cloudflare blog, 2025.
- Cloudflare audits key transparency for WhatsApp — Cloudflare blog, 2024.
- Humanity wastes about 500 years per day on CAPTCHAs — Cloudflare blog, 2021.

Open source

tlock-rs and drand-rs (timelock encryption in Rust) · age-plugin-simplepq and age-plugin-hpke (post-quantum and HPKE plugins for age) · ohttp-ts (Oblivious HTTP in TypeScript) · privacypass-ts (Privacy Pass in TypeScript) · web-bot-auth (Web Bot Auth libraries)

Education & awards

PICCASO Awards — Bridging the Privacy and Security Gap Award · 2025

Imperial College London — MSc Computing, Security and Reliability · 2017 – 2018

ENSEEIHT, Toulouse — Engineering degree, Mathematics and Computer Science · 2015 – 2018

Lycée du Parc, Lyon — Classes préparatoires, PC* · 2013 – 2015